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STEADY SEDIMENTATION AND LITHOLOGIC COMPLETENESS, BERMEJO BASIN, ARGENTINA[1]

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ABSTRACT

Late Miocene strata of the Bermejo basin, Argentina, (11–7 Ma) yield a magnetic polarity record of unusually high fidelity. Three sections, spaced along-strike over 20 km, are firmly correlated with the magnetic polarity time scale. All three sections demonstrate high accumulation rates (up to 0.88 mm/yr) and include an interval of normal polarity (extending from 7.84 to 7.81 Ma) that has yet to be resolved on the magnetic polarity time scale. The "polarity completeness" demonstrated by Bermejo strata is a result of high accumulation rates as well as environmental factors. Strata were deposited in an arid setting in which unconfined flows were the primary transporters of sediment. Shallow flow depths led to low basal shear stresses, and thus to little potential for erosion; these interpretations are corroborated by field observations. Steady sedimentation in the Bermejo basin is expressed by the low variance short-term accumulation rates have from the average long-term accumulation rate. Uniformity of sedimentation in the Bermejo basin is expressed by identical accumulation histories, even though time-transgressive formational boundaries and texture indicate that the depositional environments were not identical at all locations. These observations suggest that a change in accumulation rate does not necessarily produce a change in depositional environment and, conversely, a change in depositional environment is not necessarily accompanied by a change in accumulation rate.

INTRODUCTION

Layered sedimentary strata are the end products of a series of depositional events punctuated by intervals of nondeposition or erosion. Strata only sample a portion of geologic time, and interpretations are highly biased toward times when deposition has occurred, conceivably a minor percentage of the total time represented by the stratigraphic section (Ager 1973). A quantitative understanding of how often, and how effectively, depositional events are recorded can permit responsible interpretation of the coeval geologic history of a region based on the strata. Workers have approached this issue through studies of temporal and lithological completeness (e.g., Sadler 1981; Crowley 1984; Retallack 1986) and studies of unsteadiness and non-uniformity of sedimentation (Johnson et al. 1988; Friend et al. 1989; McRae 1990). The success of these attempts is directly related to how tight the constraints are on stratal ages.

Recent research has focused on how environmental factors at the time of deposition led to significant variation in the short-term accumulation rate through time, and spatial variation in accumulation rate for a given time (McRae et al. 1987; McRae 1990; Tauxe and Badgley 1988; Friend et al. 1989). This variation in rate is an expression in the preserved strata of the inherent unsteadiness and nonuniformity of sedimentation deposition (Friend et al. 1989). Field data from Pakistan, including erosional features such as channel cuts (Friend et al. 1989) and paleosol development (Behrensmeyer pers. comm.) support this conclusion.

Magnetic polarity stratigraphies for Miocene strata of the Bermejo basin, Argentina, also provide a secure chronologic foundation for detailed work (Johnsson et al. 1984, 1986; Reynolds 1987; Reynolds et al. unpub. data).

This paper reports the results of an along-strike study of magnetic polarity in a key 1200 m section near the base of the exposed Miocene section at Sierra de Huaco. This along-strike analysis is of a different spatial scale than similar studies completed in Pakistan (e.g., McRae et al. 1987; Tauxe and Badgley 1988). The Argentine sections are thicker (over 1000 m compared to 10s of meters) and were sampled with less detail (10–25 m compared to ~1–5 m). Furthermore, a much larger distance separates adjacent sections (10 km compared to 100s of m). The larger

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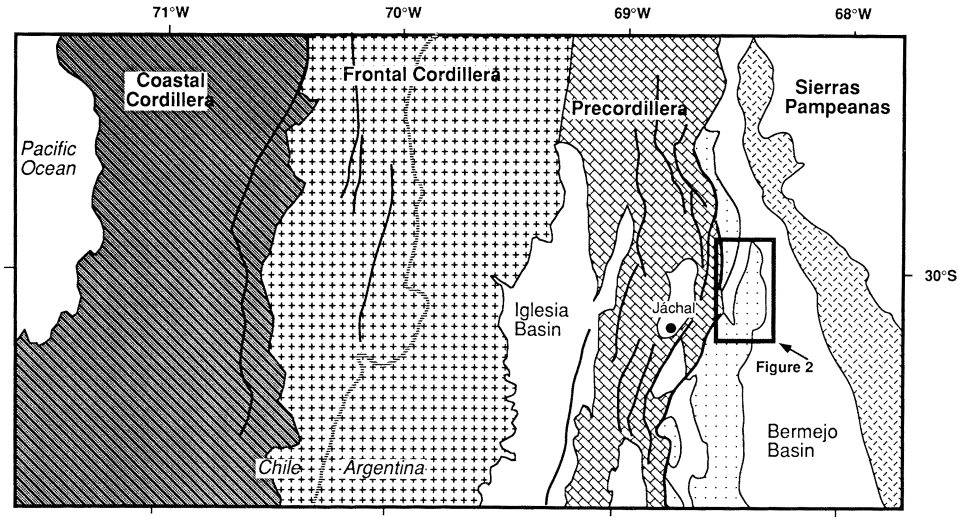


FIG. 1.—Geologic provinces at the latitude of Huaco (30°S). Boxed area is shown in figure 2.

distance between adjacent sections was chosen to maximize the possibility of observing significant differences in temporal characteristics—differences that are not implied by physical characteristics, which can be traced along strike for over 100 km. In this paper, we compare Argentine data with a Pakistani section reported in Johnson et al. (1982), which covers the same time interval and was sampled with a similar spacing between sites.

On the scale at which observations can be made using magnetic polarity stratigraphy, the Bermejo basin provides an opportunity to study steady sedimentation and a lithologically complete section. These interpretations can be substantiated using field observations, suggesting that significant insight into basic stratigraphic issues can begin with simple sedimentologic observations.

GEOLOGIC SETTING

The Bermejo basin (fig. 1) lies between the Precordillera thrust belt and the Sierra de Valle Fértil, one of the uplifts of the Sierras Pampeanas that exposes crystalline basement (Jordan and Allmendinger 1986). The main source of detritus in the Bermejo basin is the Frontal Cordillera (fig. 1), the principal topographic element at this latitude. Precordillera thrust faults are north-south trending and primarily east-verging structures that lie between the Frontal Cordillera and Bermejo basin (Ortiz and Zambrano 1981; Allmendinger et al. 1990). The thrusts deform Paleozoic and Mesozoic strata that are the major source for sedimentary lithic clasts, and the only source for limestone clasts (Furque 1979). Deformation in the Precordillera began by 16 Ma but was not initiated synchronously along strike (Beer and Jordan 1989; Reynolds et al. unpub. data; Damanti and Jordan unpub. data); subsequent thrusting in the Precordillera is thought to be episodic (Beer et al. 1990).

The Sierra de Huaco anticline (fig. 2) is located above a zone where east-verging faults of the Precordillera intersect west-verging faults associated with the Sierras Pampeanas province (Fielding and Jordan 1988; Allmendinger et al. 1990). The Sierra de Valle Fértil was not active until after 5 Ma (Reynolds 1987), and there is no evidence of its influence on Neogene sedimentary events at Huaco.

In the eastern flank of the Sierra de Huaco anticline (fig. 2), over 5.5 km of Neogene strata are exposed that can be traced along strike for nearly 100 km. The quality of these exposures is excellent, a reflection of the region's arid climate (<200 mm/yr annual precipitation). The strata at Sierra de Huaco record nonmarine deposition in the Bermejo basin from 14 to 2 Ma, coincident with crustal shortening in structural provinces to the west. The most recent phase of deformation in the Precordillera (post 2 Ma) exposes the Neogene strata at Huaco, which serve as a

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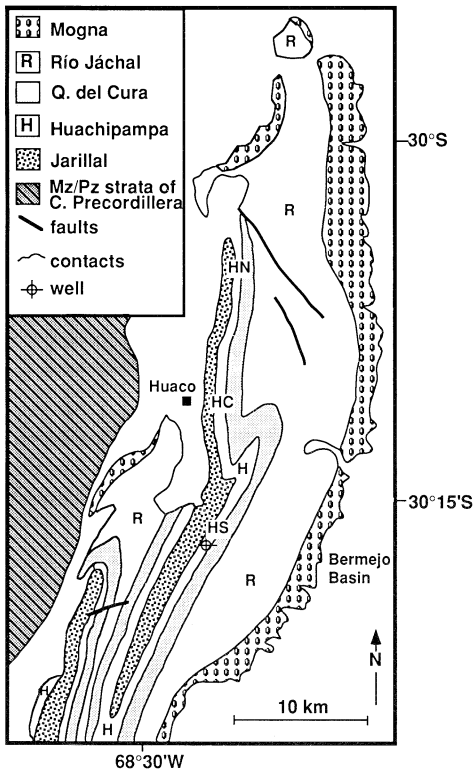


FIG. 2.—Geologic map of Sierra de Huaco anticline with the location of the three sections discussed in this paper. The sections Huaco North, Huaco Central, and Huaco South are designated by HN, HC, and HS, respectively. In the key: Mz = Mesozoic; Pz = Paleozoic; C = Central.

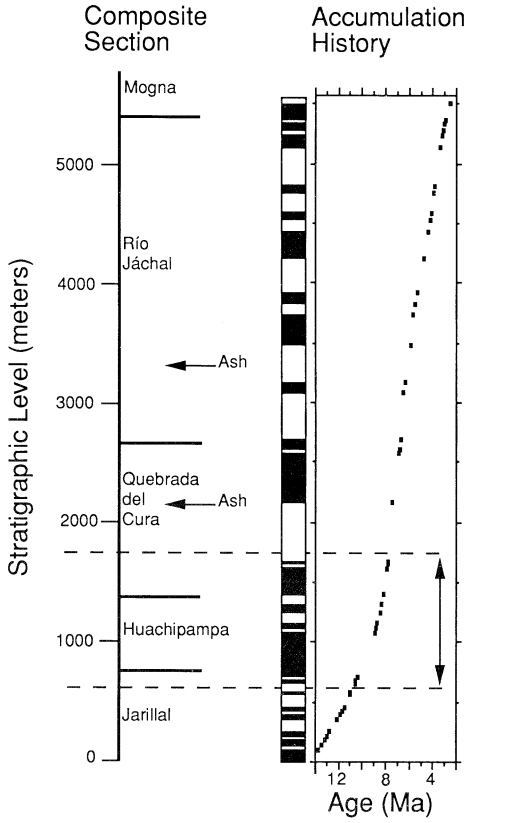


FIG. 3.—Neogene stratigraphy at Huaco. Age data from Johnsson et al. (1984) and Johnson et al. (1986). The interval studied in this paper is bracketed by dashed lines.

source for detritus in the modern Bermejo basin. From 14 to 2 Ma material deposited at the Sierra de Huaco site was transported from the west.

The oldest formation in the section, the Jarillal Fm (fig. 3), represents an ephemeral sandy braidplain whose rivers may have flowed directly from the Frontal Cordillera (Beer and Jordan 1989). The Huachipampa and Quebrada del Cura Fms and the lower Río Jáchal Fm are composed of sediment shed, in part, from Precordillera thrust plates (Johnson et al. 1986). Unchannelized flows on sandflats and distal alluvial fans were the dominant sediment transport mechanism for these formations (Beer and Jordan 1989). The upper Río Jáchal Fm and the Mogna Fm represent deposition of conglomeratic material on proximal alluvial fans (Johnson et al. 1986).

No field evidence exists for unconformities between or within the formations that comprise the Neogene section at Sierra de Huaco. This observation is supported by unpublished industry seismic data from the Bermejo basin (collected for Yacimientos Petrolíferos Fiscales, Buenos Aires), which display a concordant package of reflections representing the Tertiary section in angular unconformity with reflections representing the underlying Mesozoic section. In addition, low-altitude aerial photos (1:30,000 scale) of Sierra de Huaco strata do not show any large-scale bed termination patterns (e.g., downlap or onlap along specific stratigraphic levels).

The stratigraphic interval investigated in this paper (which includes the Jarillal, Huachipampa, and Quebrada del Cura Fms) is characterized by planar bed contacts (Huachipampa and Quebrada del Cura Fms) or

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undulating contacts that represent primary depositional relief (Jarillal Fm) (Beer and Jordan 1989). Paleosols, channel cuts, and multi-storied sandstone bodies, features generally associated with nonmarine deposits, are not observed in this portion of the section. The upper part of the Neogene sequence has not been investigated in detail, but in reconnaissance demonstrates more relief along bedding contacts and channeling on the scale of 1–5 m.

PREVIOUS WORK

Two complementary studies of the magnetic polarity stratigraphy at Sierra de Huaco provide the chronologic foundation for this research. Johnsson et al. (1984) studied the upper half of the section (encompassing the Río Jáchal and Mogna Fms) and Johnson et al. (1986) studied the lower half (encompassing the Jarillal, Huachipampa, and Quebrada del Cura Fms). The magnetic polarity sections were firmly correlated with the magnetic polarity time scale by fission-track ages of two bentonites, one from each of the two sections (fig. 3). Data from both studies successfully pass reversal tests, demonstrate a clockwise rotation, and cover a predicted time interval (12 m.y.), given the number of reversals found and an exponentially random sampling program (Johnson and McGee 1983). Magnetite was found to be the principal carrier of depositional remanence, with maghemite and titanohematite accessory carriers (Johnson 1984).

The lower half of the Sierra de Huaco section (Huaco II of Johnson et al. 1986) has been designated Huaco Central in this study (fig. 2). A 1200 m portion of Huaco Central, spanning the upper Jarillal, Huachipampa, and lower Quebrada del Cura Fms, records six intervals of normal polarity between approximately 10.7 and 7.6 Ma (fig. 3). This stratigraphic section captures the long interval of normal polarity between 10 and 9 Ma (Sea Floor Anomaly 5), as well as a short interval of normal polarity at ~7.8 Ma not included in recent compilations of the magnetic polarity time scale (Berggren et al. 1985). There is a significant increase in the sedimentary accumulation rate at ~8.7 Ma; an analysis of stratal thickness, corrected for an assumed degree of progressive compaction (Jordan et al. 1988) also shows a significant rate increase at this time. This rate increase is coincident with an important compositional change (Damanti and Jordan unpub. data), as clasts from the central Precordillera make their first appearance. These changes occur near the Huachipampa-Quebrada del Cura formational contact, which is characterized by a shift to coarser grain sizes. Beer and Jordan (1989) interpret this textural change to reflect a basinward migration of facies coincident with thrusting at ~8.7 Ma.

The older Jarillal-Huachipampa formational contact (~10.3 Ma) represents a transition to finer-grained and laterally-continuous beds and is not associated with any change in accumulation rate. Beer and Jordan (1989) believe this formational boundary to be a result of thrust activity in the western Precordillera at ~11 Ma. At that time the Huaco site may have been outside the flexural wavelength associated with structural activity in the western Precordillera. The boundary does record, however, a change in drainage patterns produced by the construction of a barrier to flow (a thrust front) and evolution of a point source (Damanti and Jordan unpub. data). Compositional changes, including an increase in the percentage of preserved sedimentary lithic clasts, are also associated with the Jarillal-Huachipampa formational boundary (Damanti and Jordan unpub. data).

The new sections described here, Huaco North and Huaco South, lie approximately 10 km north and south, respectively, of Huaco Central (fig. 2). All three sections span the same 1200 m interval and are sufficiently similar that the same formational nomenclature used at Huaco Central is appropriate.

METHODS AND RESULTS

At Huaco North and Huaco South, magnetic polarity stratigraphies were established following the sampling program and laboratory procedures of Johnson et al. (1986). The first year, a general reversal pattern was determined by collecting samples at sites separated by approximately 25 m of sediment. From each site, three oriented hand samples were extracted (details given in Johnson et al. 1975) and then analyzed at the paleomagnetic lab at Dartmouth College. A second field season was required to investigate in more detail the stratigraphic levels where reversals oc-

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cur. In total, the polarity at 112 sites was determined: 83 sites the first year (41 from HN, 42 from HS), and 29 sites the second year (16 from HN, 13 from HS), with an average site spacing of 20 m.

The natural remanence magnetization (NRM) of all the samples included a low-blocking temperature viscous remanence magnetization (VRM) component that was removed by thermal demagnetization. The characteristic component, which we interpret to be the detrital remanence magnetization (DRM), was generally revealed after thermal demagnetization of 300°C.

A blanket treatment of 500°C, the temperature utilized in previous studies at Sierra de Huaco, yielded excellent results for 90% of the samples. Almost all of the samples that required higher-temperature demagnetization (up to 650°C) are from the Jarillal Fm, where hematite is an important magnetic carrier (Johnson et al. 1986). For Jarillal samples, the characteristic magnetization was more difficult to interpret, because it represented a smaller percentage of the total magnetic vector. At Sierra de Huaco, NRM intensities range from 1×10^{-5} to 2×10^{-4} gauss (1×10^{-2} to 2×10^{-1} A/m), and post-thermal demagnetization intensities range from 1×10^{-6} to 1×10^{-5} gauss (1×10^{-3} to 1×10^{-2} A/m). In general, the magnetic behavior of Huaco North and Huaco South samples was identical to those studied at Huaco Central, and readily interpreted.

The test of randomness of vector components from multiple samples within a site (Watson 1956) indicates that 90 of the sites (83%) are statistically significant at the 95% confidence level (Class I sites, designation after Opdyke et al. 1977) and 17 of the sites (16%) are not statistically significant, but their polarity is not in doubt (Class II sites). In each Class II site, one of the three samples had a magnetic vector significantly at variance with the other two, the errant third sample always having a shallowly dipping, east-west oriented magnetic vector. Given the ambiguity associated with these vectors, the polarity assignment was made on the basis of the other two samples. Two of the sites have samples with randomly oriented magnetic vectors interpreted as the product of more severe diagenetic alteration. They are not at critical stratigraphic levels (i.e., they are in middle of a series of normal or reversed sites) and can be excluded from the data set.

Statistical analysis of Class I site vectors is presented from both Huaco North and Huaco South (table 1). The antipode of the mean of reversed site vectors is approximately coincident with the mean of normal site vectors (table 1; fig. 4), supporting the conclusion that Sierra de Huaco samples are accurate recorders of the geomagnetic reversals.

The polarity pattern for the sections is derived from the position of the virtual geomagnetic pole (VGP) latitude for each site (fig. 5). The boundary between zones of opposite polarity is taken as the point halfway between the sites of opposite polarity. All but two of the 26 polarity zones are defined by more than one site, suggesting that none of

TABLE 1

MEAN DIRECTIONS OF NORMAL AND REVERSED SITE VECTORS

	Inclination (°)	Declination (°)	N^a	R^b	R_o^c	k^b	α_{95}% (°)	α_{99}% (°)
Huaco North								
Normal	10.5	−37.9	30	28.1	8.8	14.4	7.3	9.2
Reverse	218.5	45.4	14	13.6	6.0	31.5	7.5	9.6
Huaco South								
Normal	2.6	−37.1	26	25.1	8.2	26.6	5.6	7.1
Reverse	193.2	45.4	20	17.5	7.2	8.0	12.7	16.1
Combined Data								
Normal	6.7	−37.6	56	53.0	12.1	18.1	4.6	5.8
Reverse	201.6	46.4	34	30.6	9.4	9.9	8.4	10.5

[a] Number of directions averaged.
[b] Statistical parameter of Fisher (1953).
[c] Statistical parameter of Watson (1956).

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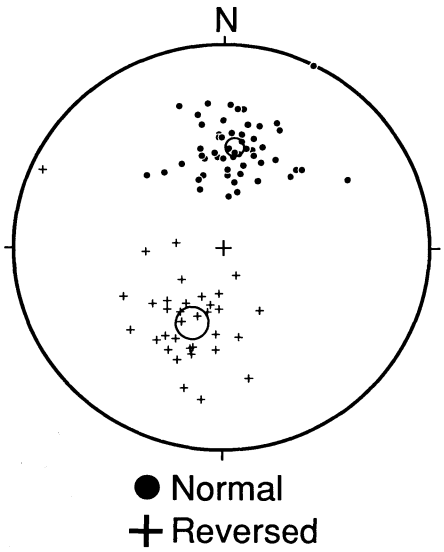


FIG. 4.—Magnetic vectors of Class I sites from Huaco North and Huaco South plotted on an equal angle stereonet. Filled circles are upper hemisphere projections, crosses are lower hemisphere projections. Open circles are cones of 95% confidence about the mean. The antipode of the mean of reversed sites is only statistically the same as the mean of normal sites at the 99% confidence level. The inability of the data to pass the reversal test at the 95% confidence level suggests that the normal overprint of reverse sites was not completely removed during thermal demagnetization.

the zones represent spurious results due to experimental errors. In both sections, six zones of normal polarity and seven zones of reverse polarity are present; included in both sections is a long zone of normal polarity that covers ~400 m of section.

The Jarillal-Huachipampa formational contact (placed arbitrarily at a stratigraphic level of 200 m) was chosen as the base line for comparison between the three sections (fig. 6). Given the criteria defining the Jarillal-Huachipampa contact (Beer and Jordan 1989), its location in the sections has an error of ±10 m.

Although there are slight differences, the polarity of Huaco North and Huaco South closely resemble the pattern at Huaco Central. The correlation of the Huaco North and South sections with the magnetic polarity time scale of Berggren et al. (1985) is based on: (1) physical stratigraphic correlation to Huaco Central, which is already correlated to the magnetic polarity time scale by ash dates (Johnson et al. 1986), and (2) direct visual comparison of Huaco North and South polarity patterns with the time scale. The correlations shown in figure 6 allow absolute age estimates to be assigned to polarity zone boundaries in the stratigraphic sections. The polarity pattern at all three Sierra de Huaco sections includes a narrow zone of normal polarity (zone N_6) that corresponds to an interval on the magnetic polarity time scale where no normal polarity interval is present.

Plots of stratigraphic position vs. age (fig. 7) highlight the chronologic similitude between the three sections. The rate of sedimentation for the lowest 500 m of section was constant and decreases from north to south (0.28 to 0.24 mm/yr) for any given time interval. At approximately 8.7 Ma, the rate increases substantially at all three sections (the points for the remainder of the curve are fitted by a third-order polynomial). By 7.9 Ma, rates had increased to 0.88 mm/yr at Huaco North, with lower values at Huaco Central and Huaco South (0.77 and 0.75 mm/yr, respectively). Rate values for both the linear and curved portions have an estimated error of ±0.02 mm/yr based on uncertainties in the exact location in the stratigraphic section of the polarity reversals that define the accumulation history (note error bars in fig. 7). Based on the experience of Jordan et al. (1988), who found that relative rates were not greatly altered by decompaction analysis, we have not manipulated the observed compacted rates. With respect to the spatial trends in accumulation rate, decompaction analysis would only emphasize the difference in rate, since more mud is present to the north than the south (discussed later).

The derived accumulation rates can be used to determine the age relationships of features common to the three stratigraphic columns but not coincident with polarity boundaries. The Jarillal-Huachipampa formational boundary, which is located within the thick zone of normal polarity, appears to be time transgressive. The boundary is older at Huaco Central (10.24 Ma) than at Huaco North (10.00 Ma) and Huaco South (9.88 Ma). If it were the same age, the accumulation rate at Huaco North and South would have been double that at Huaco Central for a short time interval, an interpretation not supported by any field or chronologic evidence.

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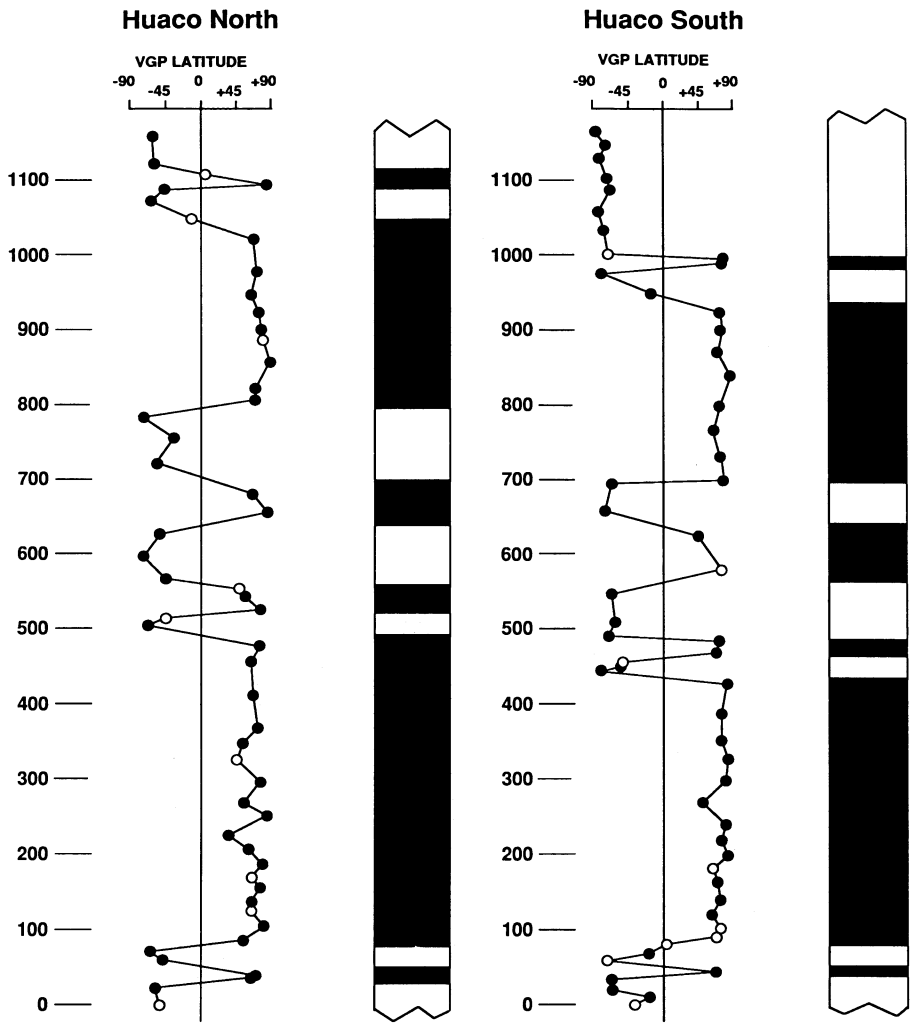


FIG. 5.—Virtual geomagnetic pole (VGP) latitudes for sites at Huaco North and Huaco South. Filled circles represent Class I data, open circles represent Class II data (see text for definitions of data type).

The Huachipampa-Quebrada del Cura contact, which is found within different polarity zones in different sections, is also time transgressive. At Huaco South it is 8.04 Ma, while at the two sections to the north it is 8.24 Ma. There is excellent agreement among the three sections for the age of the normal polarity zone at the top of the section, for which no correlative interval is found in the magnetic polarity time scale (fig. 6). This interval has a duration of approximately 30,000 years, between 7.84 and 7.81 Ma.

The texture of the three sections (fig. 8) is expressed by a sand:mud ratio and by the size of the sand fraction. The sand:mud ratio is the moving average of a window of 10 measurements through the sections. Each measurement represents a visual estimate of the percentage of sandstone over an interval, which is a straightforward process because of the sharp difference in color between the white sandstones and the dark red siltstones and claystones. The "width" of the window was approximately 100 m of section. At Huaco South, pebbly sandstones occupy specific stratigraphic zones within the upper part of the Huachipampa Fm (labeled "c" in fig. 8); when traced to the north, the pebble volume in these sands diminishes. In general, the three sections show a fining trend to the

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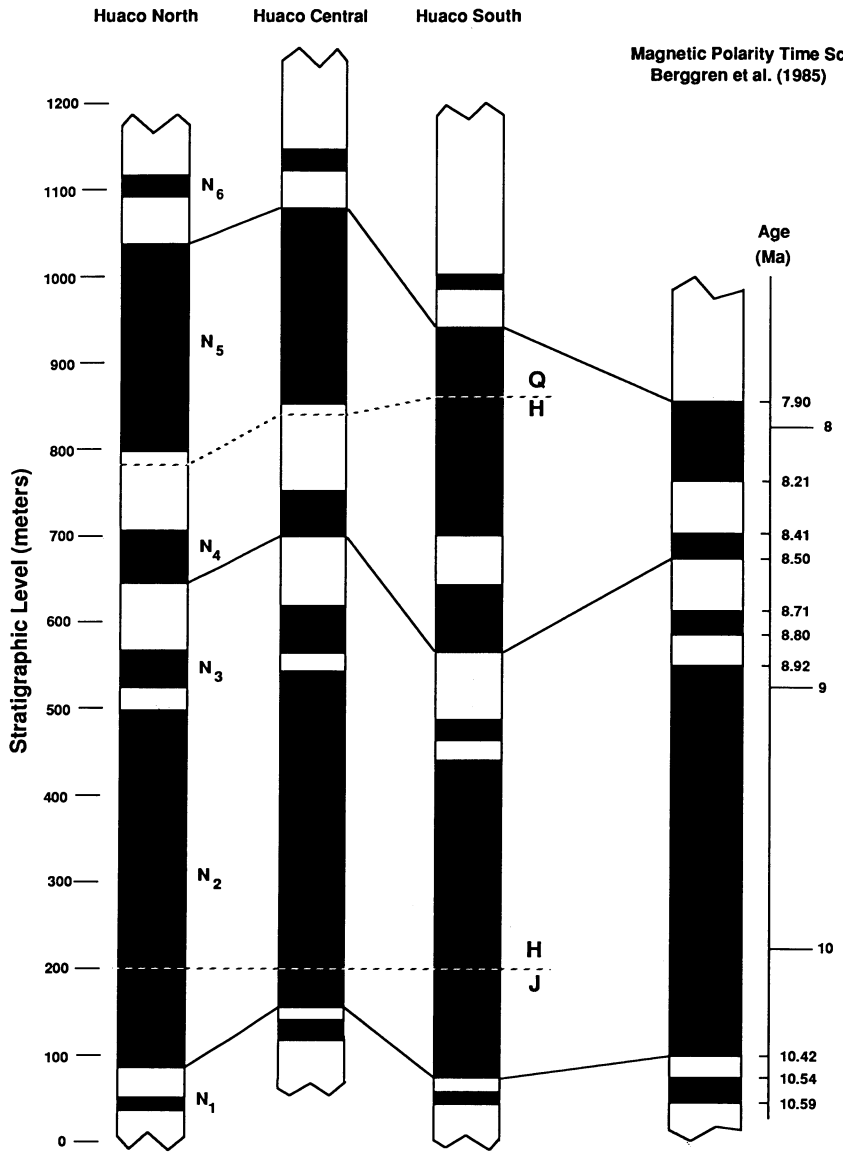


FIG. 6.—Correlation of polarity patterns at Huaco North, Central, and South with the polarity time scale of Berggren et al. (1985). Huaco Central data are from Johnson et al. (1986). Stratigraphic datum is the Jarillal-Huachipampa formational contact, which is assigned a stratigraphic level of 200 m.

north, the result of an increase in the mud proportion as well as an absolute fining of the sand fraction.

Time lines on figure 8 represent the ages of polarity boundaries that can be correlated to the magnetic polarity time scale. Time lines are generally parallel to one another, as already suggested by the similarities between the polarity patterns of the three sections. However, time lines are not parallel with respect to formational contacts, because of the time-transgressive nature of the lithostratigraphic contacts.

Time-stratigraphic columns for the three sections (fig. 9) can be constructed utilizing the same textural data displayed in figure 8 and the accumulation rate data derived from figure 7. The shift to coarser sedimentation

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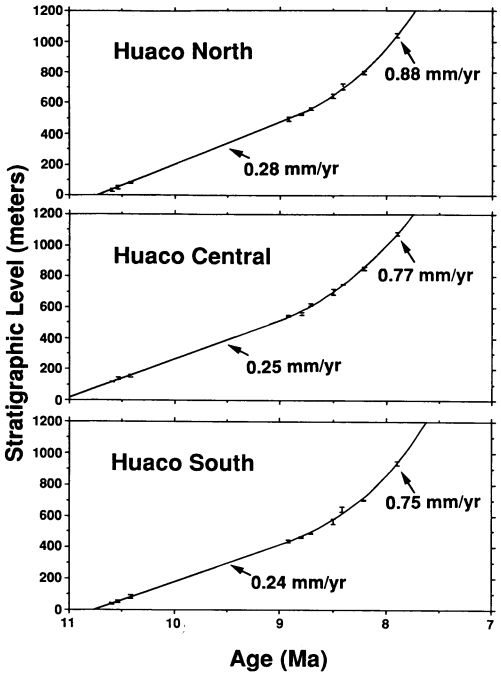


FIG. 7.—Stratigraphic level versus time for the three Sierra de Huaco sections. Slope of the line in these plots yields the sedimentary accumulation rate. All three sections demonstrate a significant increase in rate at approximately 8.7 Ma.

that occurs near the Huachipampa-Quebrada del Cura formational boundary is synchronous along strike, occurring about 8.4 Ma (corresponding to line A).

POLARITY COMPLETENESS OF THE BERMEJO BASIN RECORD

The magnetic polarity time scale provides a global standard with which different magnetic polarity sections can be compared. The "completeness" of a magnetic polarity section (here termed "polarity completeness") is defined as the percentage of reversals captured in a measured section that are included on the magnetic polarity time scale. Polarity completeness is neither temporal nor lithological completeness, but a measure of the

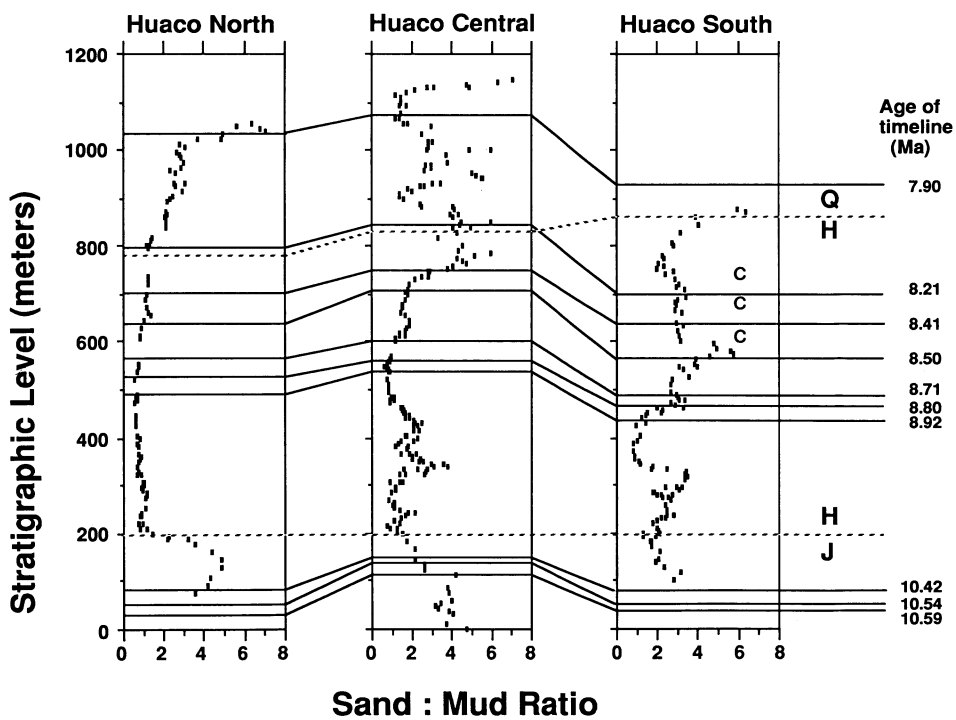


FIG. 8.—Rock-stratigraphic columns and superimposed time lines. Texture of the strata at the three sections is expressed as a sand:mud ratio ("c" refers to zones with pebbles). Formation boundaries are marked by dashed lines (J = Jarillal; H = Huachipampa; Q = Quebrada del Cura).

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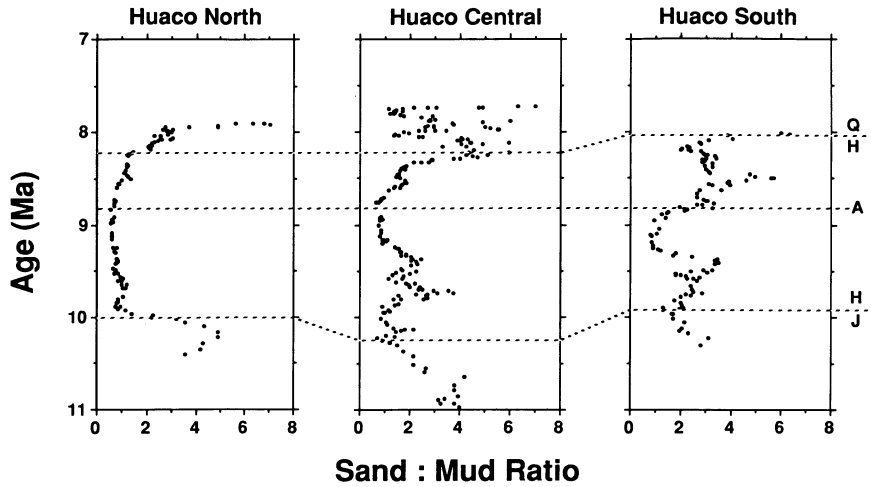


FIG. 9.—Time-stratigraphic columns showing texture and time-transgressive formational contacts. Time line "A" marks the beginning of coarser sedimentation due to thrusting in the Precordillera. Formation boundaries are marked by dashed lines (J = Jarillal; H = Huachipampa; Q = Quebrada del Cura).

ability of a sedimentary basin to preserve changes in the earth's magnetic field. Sections with high accumulation rates are apt to capture more reversals than equivalent sections with low accumulation rates. Sections with high accumulation rates may also record short-duration polarity intervals not yet included on the magnetic polarity time scale, which is based on marine magnetic surveys of finite resolution.

Details of the polarity pattern, such as the "extra" zone of normal polarity recorded at Sierra de Huaco sections (zone N_6, fig. 6), address the limits of the ability of strata to record reversals. Zone N_6 can be interpreted as an anomalous feature produced by post-depositional alteration or as the record of a polarity interval that has yet to be resolved on the magnetic polarity time scale (Berggren et al. 1985).

All of the Huaco sections have at least one Class I site that defines zone N_6. The stratigraphic interval corresponding to zone N_6 represents a sandy coarse member/fine member cycle, units typical of the Quebrada del Cura Fm (Beer and Jordan 1989). No properties suggestive of a strong diagenetic influence were observed, and the behavior of samples during thermal demagnetization was no different than that observed at other sites within the Huachipampa and Quebrada del Cura Formations.

The issue of resolution can be indirectly evaluated by noting the duration of intervals included on the Berggren et al. (1985) magnetic polarity time scale. Of the 166 intervals of normal and reverse polarity included in the Cenozoic time scale, only three intervals are of a duration equal to or less than 30,000 yr (the time represented by zone N_6 at Huaco). Thus, the limit of resolution in the marine anomaly record is of the same scale (a few 10s of thousands of years) as the Huaco interval, suggesting that zone N_6 may record a "real" polarity reversal missing from the polarity time scale.

Previous magnetic polarity time scales (e.g., Ness et al. 1980) have included intervals of short duration (<40,000 yr) not on the time scale of Berggren et al. (1985). These short-duration intervals represent short-wavelength magnetic anomalies that are superimposed on the first-order polarity pattern. Four anomalies within Chron 9 (or Anomaly 5; zone N_2, this study), found by Blakely (1974) in a statistical analysis of marine magnetic profiles, were not observed in the Huaco sections. No normal polarity interval has been proposed within the interval of reversed polarity from 7.9 to 7.41 Ma (the age of zone N_6), but the kind of detailed analysis done by Blakely (1974) has not been attempted for this interval. LaBrecque et al. (1977) have questioned whether anomalies

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such as those reported by Blakely (1974) are actual reversals of the magnetic field or simply geomagnetic field intensity fluctuations. The Huaco data do not address this issue, since short-wavelength anomalies found in the marine record were not found at Huaco, and interval N_6 at Huaco has not yet been identified as a consistent marine magnetic anomaly.

If zone N_6 does record an interval of normal polarity of the earth's magnetic field, other high-resolution sections that span the same time interval should corroborate the interpretation. At least one other magnetic polarity section has a zone of normal polarity in approximately the same temporal position (7.84 to 7.81). Butler et al. (1984), in a study of Miocene nonmarine deposits in northern Argentina (Catamarca Province), report a zone of normal polarity 180 m below the stratigraphic level corresponding to 7.41 Ma (as given by Berggren et al. 1985). Using the average accumulation rate—0.56 mm/yr—that was determined for the entire 4.5 m.y. section, the age of the zone extends from ~7.78 to 7.73 Ma. It is, however, uncertain whether this is an appropriate rate for this calculation, as the long-term rate may be inappropriate for such a short time interval. Also, the location within the section of the boundaries of the polarity zones is only constrained to ±11 m. Given these factors, the age and duration of this zone in Catamarca Province *could* be identical to that of zone N_6 at Sierra de Huaco.

Several magnetic polarity sections have been established in Pakistan that include the interval 11 to 7 Ma (Johnson et al. 1982), but no section has an interval of normal polarity in the same temporal position as that observed at Sierra de Huaco. It should be noted, however, that even the section with the closest resemblance to the magnetic time scale (the Kotal Kund section, Johnson et al. 1982) is missing polarity intervals that are included in the time scale; for instance, the interval of normal polarity between 8.80 and 8.71 Ma is not recorded, a much longer interval than that represented by zone N_6 at Huaco.

The cause for the great difference in fidelity between the sections in Pakistan and those at Sierra de Huaco may in part be related to the sedimentation rates in the two basins. Few

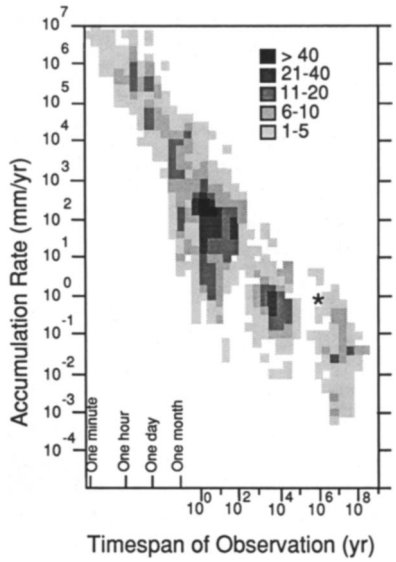


FIG. 10.—Rates of accumulation of fluvial strata for nonmarine strata plotted against the time span for which they were determined (after Sadler 1981). Key to density scale (of number of observations) given in upper right-hand corner. Sierra de Huaco sections demonstrate accumulation rates of 0.75 mm/yr for over 2.5 m.y. (marked with an asterisk), which are relatively high for this time span of observation.

nonmarine sections demonstrate long-term sedimentary accumulation rates greater than 0.75 mm/yr, the rate documented for Huaco at 7.9 Ma. This is graphically expressed in a comparison with rate data compiled by Sadler (1981) for nonmarine sections (fig. 10). The Sierra de Huaco rate, which was maintained at 0.75 mm/yr or higher for over 2.5 m.y. (Johnson et al. 1986), exceeds almost all other rate determinations for that span of observation.

Long-term rates (such as the rate of 0.75 mm/yr rate at Sierra de Huaco) are a function of the structural setting, which controls subsidence of the basin, rather than of the depositional environment (Sadler 1981; Friend et al. 1989). High accumulation rates result in a more complete magnetic polarity stratigraphy because, even if deposition is unsteady, the basin will likely receive some sediment during a polarity interval. At rates of 0.75 mm/yr, over 22 m of section can be deposited in 30,000 yr, the approximate duration of the shortest observed polarity interval. Most sampling programs (which initially may use a

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site spacing of over 10 m) can capture such an interval in a section that accumulated rapidly but may miss it in one that accumulated more slowly.

The importance of erosion within a depositional environment, another factor that affects both polarity and lithological completeness, can be evaluated through field observation. Sierra de Huaco sections do not display evidence of channelling or other features associated with erosional processes. In the Huachipampa and Quebrada del Cura Fms (which represent most of the strata measured in the magnetic polarity sections), the bases of beds are notably planar and show practically no relief on the scale of individual outcrops (Beer and Jordan 1989). Only the Jarillal Fm has channel forms, but sedimentary structures and bed architectures indicate that streams were not highly erosive (Beer and Jordan 1989).

Bed geometries and structures indicate that sheetfloods were the dominant transporter of sediment to this portion of the Bermejo basin during the interval 10 to 7 Ma (Beer and Jordan 1989). The modern Bermejo basin environment provides an excellent analogue: flows enter the basin through narrow canyons, spread out over low-gradient alluvial fans (<1° slope) and then merge with flows from adjacent fans on a vast, featureless sandflat with little relief. The potential for scour and erosion in any setting is primarily a function of the flow cross section. If all other variables are held constant, the greater the area covered by a flow, the smaller the rate of scour (Laursen 1953). In arid closed basins, where the flow cross section is essentially infinity and flow depths very shallow, little scour will occur, assuring that the next flow will also have little tendency to erode.

In a very simple system with steady, open channel flow, the relationship between erosion and flow depth is expressed quantitatively in the resistance equation (Henderson 1966). Basal shear stress (τ_o), which reflects the potential for scour, is a function of the specific weight of the fluid (γ), hydraulic radius (R), and slope (S_o).

$$\tau_o = \gamma R S_o$$

For very wide channels or, in this case, a channel of infinite width, the hydraulic mean radius (R) can be approximated by the depth of the flow. Thus, even though the flow velocity of a sheetflood may be high, the erosive power of the flow (as measured by the basal shear stress) may be relatively low. Alternatively, the short duration over which sheetflood events occur may be a factor limiting channel development. The relative importance of these controls (e.g., flow depth vs. event duration) in producing erosion has not been addressed in empirical studies of sheetflood events in arid settings.

Low erosion potential is dramatically recorded in the modern Bermejo basin (Beer and Jordan 1989). Almost no channels with relief greater than 0.5 m have been cut into the modern depositional surface, with the exception of the Bermejo River trunk stream that runs along the eastern margin of the basin. The only significant morphological relief in the basin is provided by isolated eolian sand dunes up to 2 m high and occupy areas less than 1000 m^2.

STEADINESS OF DEPOSITION IN THE BERMEJO BASIN

The steadiness of sedimentation can be evaluated on accumulation plots by observing the short-term variation in rate with respect to the long-term rate. Friend et al.'s (1989) useful method for estimating steadiness of sedimentation at the time of deposition uses preserved stratal thicknesses and precise age control, which define a "surface variability parameter" that is the mean deviation of the short-term stratigraphic level (controlled by short-term accumulation rates) from a reference level (controlled by long-term accumulation rates). Highly unsteady sedimentation will yield stratigraphic sections with large surface variability values compared to sections that have accumulated more steadily.

These concepts are illustrated in a comparison between accumulation plots from Pakistan and Huaco for the interval 11 to 7 Ma (fig. 11*A*). At Huaco, when accumulation rate was constant (and relatively slow), the short-term rate varies little from the long-term accumulation rate. From Pakistan, the Kotal Kund section (Johnson et al. 1982) covers the entire 11 to 7 Ma interval and demonstrates the most polarity completeness of all the sections studied (even though one inter-

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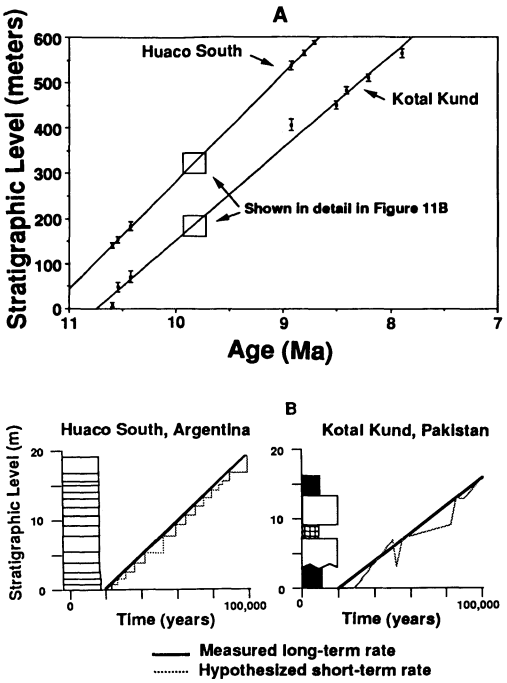


FIG. 11.—*A*. Comparison of sedimentary accumulation rates from Pakistan (Kotal Kund section, Johnson et al. 1982) and Huaco South, Argentina for the time interval 11 to 7 Ma. The absence of data points within the Pakistan data at 8.71 and 8.80 Ma indicate that a polarity zone was not observed in the section during this time. Vertical lines on data points show errors in the location of polarity boundaries. The Pakistan data demonstrate more variability of short-term accumulation rates relative to the long-term rate than do Huaco data. *B*. Hypothesized short-term accumulation rate for a 20 m stratigraphic interval from both sections. The rates for the Pakistan section follow the work of Friend et al. (1989) and McRae (1990). (Stippled pattern = sand; dark pattern = mud; checkered pattern = paleosol).

val of normal polarity from 8.80 to 8.71 is not present). At Kotal Kund there is much greater variability in the short- and long-term accumulation rates; error bars indicate that this variability is not due to poorly-constrained reversal sites. The contrast between these two sections is related to the different sedimentation processes in their respective depositional environments. Whereas Bermejo strata were deposited by sheetfloods on arid alluvial fans and sandflats/playa lakes in a closed basin (Beer and Jordan 1989) the Pakistan strata were deposited by high-discharge meandering streams that drained to the ocean (Behrensmeyer and Tauxe 1982; Johnson et al. 1985).

Miocene strata in Pakistan are characterized by three primary fluvial sub-environments, all with different accumulation rates (Friend et al. 1989; McRae 1990): channel sandstones accumulate rapidly (~100 mm/yr), overbank mudstones more slowly (~1 mm/yr), and paleosols even slower (~0.5 mm/yr). The Sierra de Huaco sections, however, are built primarily of sheet sandstones representing individual, relatively instantaneous sheetflood events. Mud in the system is mostly silt-sized material deposited lateral to and distal of their coeval sandy equivalents (Beer and Jordan 1989). Paleosol development has not been observed—notably absent are calcic horizons (immature caliches) common in other arid Miocene deposits in the Bermejo region (Beer et al. unpub. data; Reynolds et al. unpub. data). The absence of paleosol development is related to the flat morphology of the alluvial fan and sandflat in the Huaco portion of the Bermejo basin and to the high accumulation rate. Unconfined flows that transport material in the basin are able to affect large areas, even though discharge for any one flow may be relatively small. Thus few zones experience prolonged nondeposition. High accumulation rates also assure that sediment consistently reaches depositional sites; Crowley's (1984) model suggests that sections with high accumulation rates preserve the products of high frequency, low-magnitude events, circumventing the normal "low-pass" filter that operates in systems with lower accumulation rates.

Another consideration in the analysis of short-term rate variability is erosion, which removes material that would have recorded deposition at the depositional site. For reasons discussed earlier, erosion is not an important factor within the Bermejo basin. The accumulation history at Sierra de Huaco can be considered as a series of sediment pulses, with little or no intervening erosion. Throughout the basin, pulses follow one another at regular intervals, leading to a continuous stair-stepped pattern, with little "relief" between stairs (fig. 11 *B;* in Friend et al. 1989, referred to as "episodic sedimentation with an equal time gap assumption"). This contrasts dramatically with similar plots for sedimentation history in Pakistan (Friend et

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al. 1989), which show significant intervals of nondeposition (and paleosol formation) as well as erosion due to channelization (fig. 11*B*).

The implication from field observations is that the sections measured at Sierra de Huaco preserve nearly all of the sediment deposited at this site. An inference from the magnetic polarity stratigraphy is that sediment accumulated in a relatively steady fashion on the time scale of 10^4 to 10^6 Ma. Interestingly, the arid environment is characterized by flash floods, a highly unsteady transporter of sediment if considered on the scale of human observation. However, on the scale of magnetic polarity stratigraphy, which is the most precise method yet developed to date strata, the accumulation history can be termed steady.

If little material was eroded between depositional events, an estimate may be made for the recurrence interval of sheetflood flows that delivered material to the Sierra de Huaco site (particularly for the Huachipampa and Quebrada del Cura Fms). For any 10 m stratigraphic interval (representing ~15,000 yr), there are 30 to 300 individual beds, of which approximately 90% were the product of sheetfloods. Assuming that deposition of sandstone beds was relatively instantaneous, approximately 50 to 500 yr elapsed between deposition of individual sheetflood beds at Sierra de Huaco. This figure may be considered the flash flood recurrence interval for *major* flows, flows that reached the Bermejo basin still bearing sand-size sediment load. Such a recurrence interval is normal for major floods in other hyper-arid areas (Baker 1977) and is much smaller than the time resolution available in magnetic polarity stratigraphy (i.e., steady sedimentation will be inferred using magnetic methods). Unconfined flows are typically short-lived events (on the scale of hours) associated with convective thunderstorm systems (Rahn 1967). The temporal completeness of the Bermejo section is thus extremely small: one day of sedimentation in 50 to 500 yr, or less than 0.001 to 0.01%!

Of what use is the knowledge that arid environmental conditions can produce a rock record that is lithologically complete, and that sedimentation was steady? The Huaco sections are of little value to paleontologists trying to document evolutionary change, because the fossil record there is nearly nonexistent due to the hostile Neogene Bermejo environment. However, arid-environment deposits may be useful in the study of time intervals such as the Cretaceous-Tertiary boundary, where an estimation of completeness will influence interpretation of major geologic events (Dingus 1984). The "capture" of a short-duration interval of normal polarity in the Sierra de Huaco sections suggests that arid-environment deposits could also be used better to define the magnetic polarity time scale. A detailed sampling program (site spacings on the order of 1 to 5 m) would probably reveal several polarity zones representing time intervals <40,000 yr, polarity intervals that cannot be resolved on marine surveys of magnetic anomalies.

The corroboration of field and magnetic polarity data suggest that bed geometries and primary sedimentary structures are useful indicators of lithologic completeness. Such field observations can guide interpretation of progressive changes through a section of some parameter of interest (e.g., clast composition or texture). If the field relationships suggest that there is a representative sampling of depositional events (and by extension, a representative sampling of source-area evolution), one may confidently link observed trends in a sedimentary section to the coeval structural development marginal to the basin.

UNIFORMITY OF SEDIMENTATION IN THE BERMEJO BASIN

The time lines that link the three sections at Sierra de Huaco provide a means to evaluate the uniformity of sedimentation in the Bermejo basin. Uniformity can be considered at many different scales, from individual beds to entire formations, and for a host of parameters (e.g., accumulation rate, sedimentary processes, and texture). The most striking feature of the Huaco magnetic polarity data is the along-strike similarity in form of the sedimentary accumulation history as well as the magnitude of rates (fig. 7). This uniformity is especially striking when compared to data from Pakistan (Johnson et al. 1988), which show much more marked lateral vari-

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ability in accumulation rate over distances much smaller than those encountered at Huaco. We interpret uniformity in long-term accumulation rate to reflect primarily on the configuration of structures on the basin margin, rather than on depositional processes (see also Friend et al. 1989).

The time-transgressive nature of the formational boundaries (fig. 8) indicates that, in terms of depositional environment, sedimentation was not uniform. The Jarillal-Huachipampa formational contact has an age of 10.24 Ma at Huaco Central and is approximately 300,000 yr younger at both Huaco North and Huaco South. The change to Huachipampa conditions is interpreted to correspond to construction of a barrier to flow in the western Precordillera that funneled detritus through a single opening, or point source (Damanti and Jordan unpub. data). This passage through the Precordillera is the valley of the Río Jáchal (fig. 1), an antecedent stream that has controlled depositional patterns in this region since the Miocene (Damanti and Jordan unpub. data). The time-transgressive nature of the contact at Sierra de Huaco may relate to the irregular geometry of the front of the alluvial fan/sandflat complex that deposited the sediment of the Huachipampa Fm. This interpretation would imply that an unconformity representing 300,000 yr exists undetected at Huaco North and Huaco South.

The other formational contact, that between the Huachipampa and Quebrada del Cura Fms, represents a migration of facies across the Sierra de Huaco site between 8.04 and 8.24 Ma. The facies front first arrived at Huaco North and Huaco Central, and then 200,000 yr later at Huaco South. This basinward migration of facies was interpreted by Beer and Jordan (1989) to be in response to thrusting in the Precordillera. Burbank et al. (1988) also describe a basinward migration of facies coeval with Neogene thrusting in Pakistan.

At Sierra de Huaco, the facies evidence, considered alone, suggests that deformation at ~8.7 Ma occurred first in the north and migrated to the south. Changes in sandstone clast composition also show a north to south trend: the first limestone clasts arrived at Huaco North at 9.3 Ma, at Huaco Central at 9.1 Ma and at Huaco South after 8 Ma (J. Damanti, pers. comm.). Changes in sedimentary accumulation rate, however, do not show similar north to south trends. The accumulation rate changes first at Huaco Central (8.8 Ma) and then slightly later at Huaco North and South (8.7 Ma). Given the inconsistency of these various indicators, it is difficult to interpret with any certainty the subtle details of structural activity in the Precordillera.

In spite of the uniformity in accumulation rate at Huaco, the formational boundaries are decidedly time-transgressive. This relationship suggests that changes in long-term accumulation rate are not produced by changes in depositional environment (Sadler 1981; Beer and Jordan 1989; Friend et al. 1989). At Huaco, this conclusion was reached by Beer and Jordan (1989) based on the observation at Huaco Central that changes in environment and accumulation rate were not contemporaneous. Not only do the other two sections individually support this finding, but the along-strike time lines provide evidence in the spatial domain that average accumulation rate and environment have no mutual dependence.

CONCLUSIONS

An along-strike study of a key Late Miocene magnetic polarity interval (11 Ma to 7 Ma) from the Bermejo basin, Argentina, demonstrates remarkable fidelity of the stratigraphic record. The three magnetic polarity sections at Sierra de Huaco all include a zone of approximately 30,000 yr (between 7.84 and 7.81 Ma) of normal polarity that cannot be correlated with magnetic polarity time scales. A similar zone has been documented at apparently the same temporal position in at least one other nonmarine section. This interval is interpreted as a geomagnetic normal polarity interval not yet included on magnetic polarity time scales compiled primarily from analysis of marine magnetic anomalies.

The "polarity completeness" demonstrated by Sierra de Huaco sections is explained by the relatively high rate at which sediment accumulated (up to 0.88 mm/yr) and the diminished influence erosional processes have in the Bermejo basin. Lack of erosion is attributed to the flat morphology of the depositional plain

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and associated shallow flow depths which yield low basal shear stresses. Field data and the polarity stratigraphy both imply that the record at Sierra de Huaco is lithologically complete.

Steadiness of sedimentation is suggested by accumulation rate plots which show little variance of the long-term and short-term rates of accumulation. This interpretation is corroborated by the absence of paleosol horizons and the absence of evidence for erosion (such as channel forms), two features associated with unsteady sedimentation. Uniformity of sedimentation is expressed by the constant form of accumulation rate history and the near identical magnitude of rates over 20 km of along-strike exposure. However, in terms of depositional processes, time-transgressive formational boundaries indicate significant lateral variability in environment. The uniformity in accumulation rate and the nonuniformity in depositional environment point to the independent nature of these two facets of the stratigraphic record.

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